

DIGITAL SAT MATH PRACTICE

Radian and Degree Conversion — SOLUTIONS

COMPLETE SOLUTIONS WITH TRAP ALERTS AND STRATEGY

Question 1 [Easy]

Correct Answer: B

Step by Step:

Step 1: Use the conversion formula: radians = degrees $\times (\pi/180)$.

Step 2: $150 \times (\pi/180) = 150\pi/180$.

Step 3: Simplify: $150\pi/180 = 5\pi/6$.

Why Each Wrong Option Fails:

Option A: $2\pi/3 = 120^\circ$. This is the result of using 120 instead of 150.

Option C: $3\pi/4 = 135^\circ$. This is the result of using 135 instead of 150.

Option D: $7\pi/6 = 210^\circ$. This mixes up adding an extra 60° .

■ **TRAP ALERT:** Students sometimes confuse 150° with nearby benchmark angles like 120° or 135° .

■ **KEY INSIGHT:** Memorize benchmarks: $30^\circ = \pi/6$, $45^\circ = \pi/4$, $60^\circ = \pi/3$, $90^\circ = \pi/2$, $180^\circ = \pi$.

■ **FAST METHOD:** $150 = 5 \times 30$, and $30^\circ = \pi/6$, so $150^\circ = 5(\pi/6) = 5\pi/6$.

■  **VERDICT:** Hand solve is faster. This is a 10 second mental math problem with benchmarks.

Question 2 [Easy]

Correct Answer: B

Step by Step:

Step 1: Use the conversion formula: degrees = radians $\times (180/\pi)$.

Step 2: $(3\pi/4) \times (180/\pi) = 3 \times 180/4$.

Step 3: $= 540/4 = 135^\circ$.

Why Each Wrong Option Fails:

Option A: $120^\circ = 2\pi/3$. Comes from mistakenly dividing by 3 instead of 4.

Option C: $150^\circ = 5\pi/6$. A common benchmark mix up.

Option D: $225^\circ = 5\pi/4$. Comes from adding instead of multiplying correctly.

■ **TRAP ALERT:** The π symbols cancel. Students who forget to cancel end up with π in their answer.

■ **KEY INSIGHT:** $\pi/4 = 45^\circ$, so $3\pi/4 = 3 \times 45^\circ = 135^\circ$.

■ **FAST METHOD:** Multiply the coefficient by the unit: $3 \times 45 = 135$.

■  **VERDICT:** Hand solve is faster. Direct benchmark multiplication takes under 10 seconds.

Question 3 [Easy]

Correct Answer: C

Step by Step:

Step 1: $270 \times (\pi/180) = 270\pi/180$.

Step 2: Simplify by dividing numerator and denominator by 90: $3\pi/2$.

Why Each Wrong Option Fails:

Option A: $\pi/2 = 90^\circ$. This is one quarter turn, not three quarters.

Option B: $\pi = 180^\circ$. This is a half turn.

Option D: $2\pi = 360^\circ$. This is a full revolution.

■ **TRAP ALERT:** Students who rush may grab $\pi/2$ thinking "270 has a 9 and $90 = \pi/2$."

■ **KEY INSIGHT:** 270° is three quarter turns. A full turn = 2π , so $(3/4)(2\pi) = 3\pi/2$.

■ **FAST METHOD:** Think in quarter turns: $270/360 = 3/4$ of a full circle = $(3/4)(2\pi) = 3\pi/2$.

■  **VERDICT:** Hand solve is faster. This is a benchmark angle everyone should know instantly.

Question 4 [Medium]

Correct Answer: A

Step by Step:

Step 1: degrees = radians $\times (180/\pi)$.

Step 2: $(7\pi/5) \times (180/\pi) = 7 \times 180/5$.

Step 3: $= 7 \times 36 = 252^\circ$.

Why Each Wrong Option Fails:

Option B: 248 is close but results from an arithmetic slip ($7 \times 35.4\dots$).

Option C: $256 = 7 \times 36.57\dots$ results from using $256/7$ incorrectly.

Option D: 244 is a distractor designed to catch those who subtract instead of multiply.

■ **TRAP ALERT:** 7×36 must be computed carefully. Students often miscompute as $7 \times 35 = 245$.

■ **KEY INSIGHT:** $180/5 = 36$. Always simplify 180/denominator first, then multiply.

■ **FAST METHOD:** Simplify first: $180 \div 5 = 36$, then $7 \times 36 = 252$.

■  **VERDICT:** Hand solve is faster. Simple division then multiplication. Desmos adds no value.

Question 5 [Medium]

Correct Answer: B

Step by Step:

Step 1: One full rotation = 2π radians.

Step 2: 2.5 rotations = $2.5 \times 2\pi = 5\pi$ radians.

Why Each Wrong Option Fails:

Option A: $4\pi = 2 \times 2\pi$. This represents exactly 2 rotations, not 2.5.

Option C: 2.5π comes from forgetting to multiply by 2π and using just π .

Option D: 10π comes from multiplying $2.5 \times 2\pi \times 2$, doubling the answer.

■ **TRAP ALERT:** Choice C is the classic trap: multiplying by π instead of 2π per rotation.

■ **KEY INSIGHT:** 1 rotation = 2π (not π). Always multiply number of rotations by 2π .

■ **FAST METHOD:** $2.5 \times 2 = 5$, so 2.5 rotations = 5π radians.

■  **VERDICT:** Hand solve is faster. Pure mental math once you know 1 rotation = 2π .

Question 6 [Medium]

Correct Answer: 8π

Step by Step:

Step 1: The minute hand completes 1 full revolution (2π radians) every 60 minutes = 1 hour.

Step 2: In 4 hours, the minute hand completes 4 full revolutions.

Step 3: Total angle = $4 \times 2\pi = 8\pi$ radians.

Step 4: As a decimal: $8\pi \approx 25.13$. Enter 8π or 25.13.

■ **TRAP ALERT:** Some students confuse the minute hand with the hour hand. The hour hand moves $1/12$ of a revolution per hour, but the minute hand completes a full revolution each hour.

■ **KEY INSIGHT:** Minute hand: 2π per hour. Hour hand: $\pi/6$ per hour. Read the problem carefully.

■ **FAST METHOD:** $4 \text{ hours} \times 2\pi/\text{hour} = 8\pi$.

■  **VERDICT:** Hand solve is faster. One line of reasoning. No calculator needed.

Question 7 [Medium]

Correct Answer: B

Step by Step:

Step 1: A reference angle of $\pi/6 = 30^\circ$.

Step 2: In Quadrant III, the angle = $180^\circ + \text{reference angle}$.

Step 3: $\theta = 180^\circ + 30^\circ = 210^\circ$.

Why Each Wrong Option Fails:

Option A: $150^\circ = 180^\circ - 30^\circ$, which places the terminal side in Quadrant II, not III.

Option C: $240^\circ = 180^\circ + 60^\circ$. This uses a reference angle of 60° instead of 30° .

Option D: $330^\circ = 360^\circ - 30^\circ$, which places the terminal side in Quadrant IV.

■ **TRAP ALERT:** Choice A (150°) subtracts instead of adds. Quadrant III means ADD to 180° .

■ **KEY INSIGHT:** QI: θ_{ref} . QII: $180 - \theta_{\text{ref}}$. QIII: $180 + \theta_{\text{ref}}$. QIV: $360 - \theta_{\text{ref}}$.

■ **FAST METHOD:** Convert $\pi/6$ to 30° , then QIII = $180 + 30 = 210^\circ$.

■  **VERDICT:** Hand solve is faster. Benchmark + quadrant formula = instant answer.

Question 8 [Medium]

Correct Answer: 40

Step by Step:

Step 1: degrees = $(2\pi/9) \times (180/\pi)$.

Step 2: $= 2 \times 180/9 = 360/9 = 40^\circ$.

■ **TRAP ALERT:** Students may try to approximate $2\pi/9$ as a decimal first and introduce rounding errors. Work symbolically instead.

■ **KEY INSIGHT:** Cancel π first, then simplify the fraction. $2 \times 180 = 360$, and $360/9 = 40$.

■ **FAST METHOD:** $180/9 = 20$, then $2 \times 20 = 40$.

■  **VERDICT:** Hand solve is faster. Clean arithmetic once π cancels. Enter 40.

Question 9 [Medium]

Correct Answer: B

Step by Step:

Step 1: radians = degrees $\times (\pi/180)$.

Step 2: $225 \times (\pi/180) = 225\pi/180$.

Step 3: Simplify by dividing by 45: $5\pi/4$.

Step 4: Note: the radius is extra information. The conversion depends only on the angle.

Why Each Wrong Option Fails:

Option A: $3\pi/4 = 135^\circ$. Subtracting 90° by mistake.

Option C: $7\pi/4 = 315^\circ$. Adding 90° instead of converting directly.

Option D: $5\pi/3 = 300^\circ$. Wrong denominator from incorrect simplification.

■ **TRAP ALERT:** The radius (12 cm) is deliberately included as irrelevant information to distract.

■ **KEY INSIGHT:** Radian–degree conversion depends only on the angle, never on the radius.

■ **FAST METHOD:** $225 = 5 \times 45$, and $45^\circ = \pi/4$, so $225^\circ = 5\pi/4$.

■ **VERDICT:** Hand solve is faster. Recognize $225 = 5 \times 45$, done in seconds.

Question 10 [Hard]

Correct Answer: D

Step by Step:

Step 1: Convert $11\pi/6$ to degrees: $(11\pi/6) \times (180/\pi)$.

Step 2: The π cancels: $11 \times 180/6 = 11 \times 30 = 330^\circ$.

Why Each Wrong Option Fails:

Option A: $300^\circ = 5\pi/3$. Uses 10×30 instead of 11×30 (drops the 1 from 11).

Option B: $315^\circ = 7\pi/4$. Confuses $\pi/6$ with $\pi/4$ and miscounts.

Option C: $324^\circ = 9\pi/5$. Results from dividing 180 by 5 instead of 6.

■ **TRAP ALERT:** 11×30 requires careful arithmetic. Students may miscalculate as $10 \times 30 = 300$.

■ **KEY INSIGHT:** $180/6 = 30$. Multiply the numerator (11) by 30 to get 330.

■ **FAST METHOD:** $11\pi/6$: denominator 6 gives $180/6 = 30$, then $11 \times 30 = 330^\circ$.

■ **VERDICT:** Hand solve is faster. One division, one multiplication.

Question 11 [Hard]

Correct Answer: 40.1

Step by Step:

Step 1: Total swing = $2 \times 0.35 = 0.70$ radians (it swings 0.35 on each side).

Step 2: Convert to degrees: $0.70 \times (180/\pi)$.

Step 3: $= 0.70 \times 57.2958... = 40.107...^\circ$.

Step 4: Rounded to the nearest tenth: 40.1° .

■ **TRAP ALERT:** Students may convert 0.35 radians to degrees and forget to double it for the TOTAL swing.

■ **KEY INSIGHT:** "Each side" means the total angle is $2 \times$ the given angle. Read carefully.

■ **FAST METHOD:** Double first (0.70), then multiply by $180/\pi \approx 57.296$. $0.70 \times 57.296 = 40.1$.

■🏠■ **VERDICT:** Desmos is faster here. The decimal multiplication $0.70 \times (180/\pi)$ is messy by hand.

Question 12 [Hard]

Correct Answer: B

Step by Step:

Step 1: Convert $11\pi/12$ to degrees: $(11\pi/12) \times (180/\pi) = 11 \times 15 = 165^\circ$.

Step 2: First satellite: 165° . Second satellite: 168° .

Step 3: $168 - 165 = 3$. The second satellite swept 3° more.

Why Each Wrong Option Fails:

Option A: The first satellite swept 165° , which is LESS than 168° , not more.

Option C: 15° is the value of $\pi/12$ (one twelfth of 180°), not the difference between the two angles.

Option D: $165^\circ \neq 168^\circ$. The angles are close but not equal.

■ **TRAP ALERT:** The question format makes students assume the first quantity is larger. Always compute before comparing.

■ **KEY INSIGHT:** Convert everything to the same unit (degrees) before comparing.

■ **FAST METHOD:** $180/12 = 15$, so $11\pi/12 = 165^\circ$. Compare to 168° . Difference = 3° .

■🏠■ **VERDICT:** Hand solve is faster. Clean fraction simplification, then subtraction.

Question 13 [Hard]

Correct Answer: D

Step by Step:

Step 1: Substitute $t = 5$: the argument = $\pi(5)/3 = 5\pi/3$ radians.

Step 2: Convert to degrees: $(5\pi/3) \times (180/\pi) = 5 \times 60 = 300^\circ$.

Why Each Wrong Option Fails:

Option A: $60^\circ = \pi/3$. This uses $t = 1$, not $t = 5$.

Option B: $150^\circ = 5\pi/6$. This divides by 6 instead of 3.

Option C: $240^\circ = 4\pi/3$. This uses $t = 4$, not $t = 5$.

■ **TRAP ALERT:** Students may forget to substitute $t = 5$ into $\pi t/3$ and just convert $\pi/3$.

■ **KEY INSIGHT:** Substitute first, then convert. $\pi t/3$ at $t = 5$ gives $5\pi/3$, not $\pi/3$.

■ **FAST METHOD:** $5\pi/3$: the denominator 3 gives $180/3 = 60$, so $5 \times 60 = 300^\circ$.

■🏠■ **VERDICT:** Hand solve is faster. Substitute and convert in two quick steps.

Question 14 [Hard]

Correct Answer: B

Step by Step:

Step 1: Supplementary angles sum to 180° .

Step 2: Convert the radian angle to degrees: $(k\pi/6) \times (180/\pi) = 180k/6 = 30k$ degrees.

Step 3: Set up equation: $(3k + 15) + 30k = 180$.

Step 4: $33k + 15 = 180$.

Step 5: $33k = 165$.

Step 6: $k = 5$.

Why Each Wrong Option Fails:

Option A: $k = 3$ gives angles of 24° and 90° , which sum to $114^\circ \neq 180^\circ$.

Option C: $k = 10$ gives angles of 45° and 300° , which sum to $345^\circ \neq 180^\circ$.

Option D: $k = 15$ gives angles of 60° and 450° , far exceeding 180° .

■ **TRAP ALERT:** Students who forget to convert radians to degrees before adding will set up the equation incorrectly.

■ **KEY INSIGHT:** Convert everything to the same unit (degrees) before setting up the supplementary equation.

■ **FAST METHOD:** $\pi/6$ in degrees = 30° , so the second angle is $30k^\circ$. Then $3k + 15 + 30k = 180$.

■  **VERDICT:** Hand solve is faster. Set up equation, combine like terms, solve in 30 seconds.

Question 15 [Hard]

Correct Answer: A

Step by Step:

Step 1: One full revolution = 2π radians in 90 seconds.

Step 2: In 25 seconds, the fraction of the revolution completed = $25/90 = 5/18$.

Step 3: Radians turned = $(5/18) \times 2\pi = 10\pi/18 = 5\pi/9$.

Why Each Wrong Option Fails:

Option B: $\pi/5$ is a distractor from confusing 25 seconds with a denominator.

Option C: $2\pi/9$ uses $2\pi/9$ as if only 10 seconds passed ($10/90 \times \pi$, incorrectly).

Option D: $5\pi/18$ comes from forgetting to multiply the fraction by 2π and using just π .

■ **TRAP ALERT:** Choice D ($5\pi/18$) uses π instead of 2π per revolution. Always remember one revolution = 2π , not π .

■ **KEY INSIGHT:** Fraction of revolution $\times 2\pi$ = radians swept. The 2 in 2π is critical.

■ **FAST METHOD:** $25/90 = 5/18$. Multiply by 2π : $10\pi/18 = 5\pi/9$.

■  **VERDICT:** Hand solve is faster. Simple fraction work, no calculator needed.

TOPIC SUMMARY: RADIAN – DEGREE CONVERSION

- Core Formula: Degrees = Radians $\times (180/\pi)$ and Radians = Degrees $\times (\pi/180)$.
- Benchmark Angles to Memorize: $30^\circ = \pi/6$, $45^\circ = \pi/4$, $60^\circ = \pi/3$, $90^\circ = \pi/2$, $180^\circ = \pi$, $360^\circ = 2\pi$.
- One Full Revolution = 2π radians = 360° . Never confuse with $\pi = 180^\circ$ for partial revolution problems.
- Quadrant Formulas for Reference Angles: QI = θ , QII = $180 - \theta$, QIII = $180 + \theta$, QIV = $360 - \theta$.
- Irrelevant Information Trap: Radius, arc length, and other measurements may appear in a problem that is really just a conversion. Stay focused.
- Cancel π first in symbolic expressions before doing arithmetic. This avoids decimal errors.
- Supplementary angles sum to 180° . Make sure both angles are in the same unit before adding.
- Most radian–degree problems on the DSAT are faster by hand than by calculator. Know your benchmarks and you'll save time.